

FORCES



COMPRESSING MATTER

INTRODUCTION



- In this universe, everything exists as matter or as energy
 - Allah Ta'ala is the creator; he has created everything in the very beginning.
 - He is the creator of matter and its form. He states
 - He is Allah, the creator, the one who gives form
 - He has also given some objects a certain degree of elasticity to withstand certain amount of force
 - For instance human skin and muscles
 - Allah Ta'ala states;
- And provide you the best forms (shapes)

Forces acting on solids



- Forces can make an object to stretch, squash, bend or twist. for example deforming a elastic material (rubber)



Cont'd



- Bungee jumpers rely on the springiness of rubber rope.
- If the rope became permanently stretched, they would stop suddenly at the bottom of their fall, rather than bouncing up and down and gradually coming to a halt.



Cont'd



- Also when a football is kicked, it is compressed for a short while, then springs back to its original shape as it pushes itself off the foot of the player



Cont'd (note)

- Ductility is a ability of a material to deform under tensile stress i.e ability to be stretched into wire
- Brittle – ability of a material to crack or shatter upon impact of a force
- Elasticity- ability of a material to return to its original shape after it has undergone stretch
- Stiffness is the ability of a material to resist deformation
- Malleability is the ability of a material to deform under compressive stress. i.e ability of a material to form a thin sheet by hammering or rolling.
- You must have seen *ayaan* engraved around the *qibla* in *masajid* in gold. Gold is moulded and transformed into thin leaves due to its malleability

Stretching springs



- Springs are designed to stretch when small force is applied
- The length of the spring increases as the applied force increases
- When the force causing the spring to stretch is withdrawn, then the spring returns to its original length
- If the load is increase such that it causes the spring not to return to its original length after the load has been removed, therefore the spring becomes permanently stretched

Cont'd

- Such a spring is said to be permanently deformed since its Elastic limit has been exceeded

Law of elasticity (*Hook's law*)

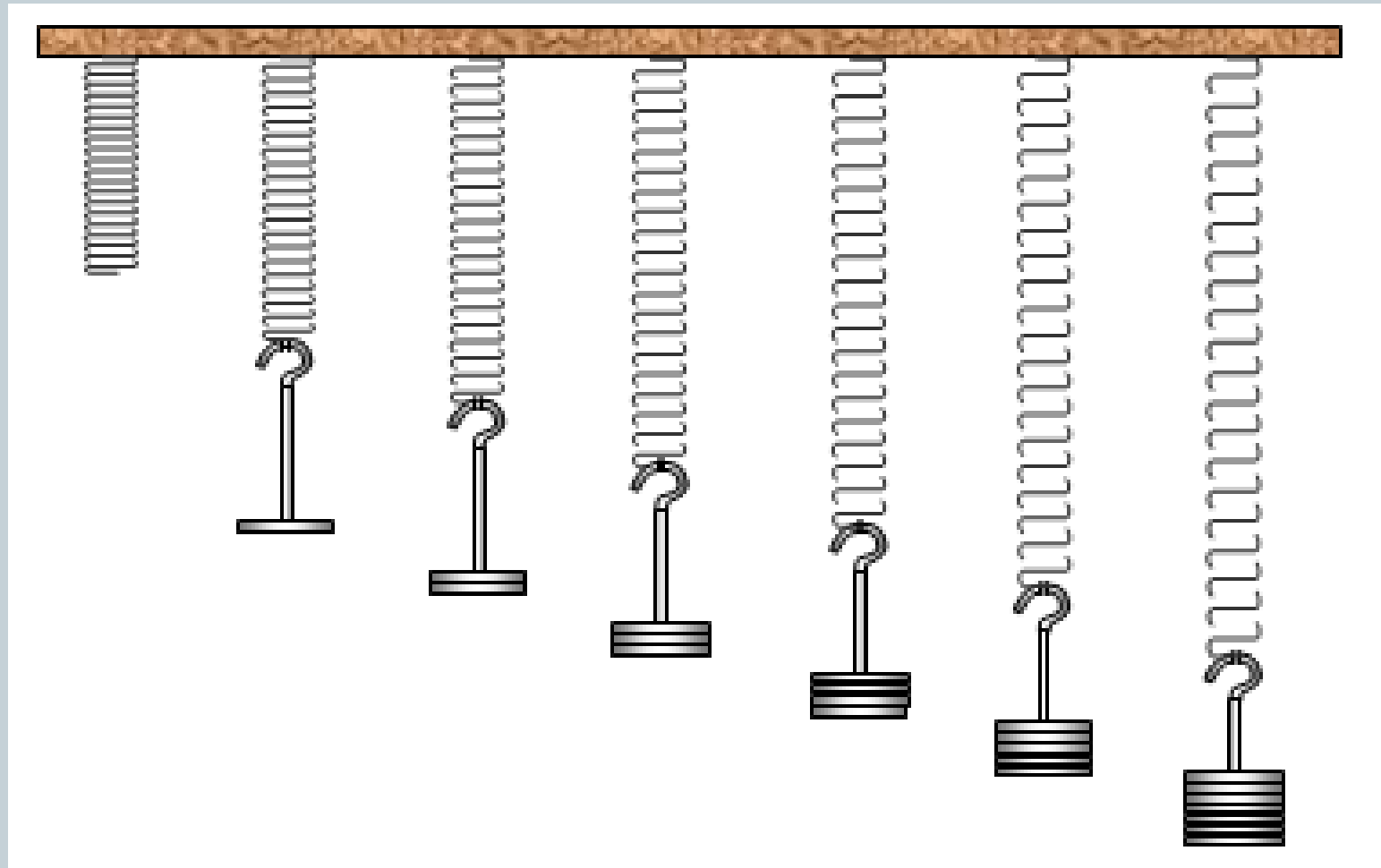
It states that

The extension of a spring is proportional to the load(force) applied to it, provided the elastic limit is not exceeded

$$F = -kx$$

- Where F is Force (load), ***k*** is the stiffness of the spring known as the *spring constant* (expressed in N/m) and ***x*** is the length of extension/compression of the spring

Cont'd



Experiment

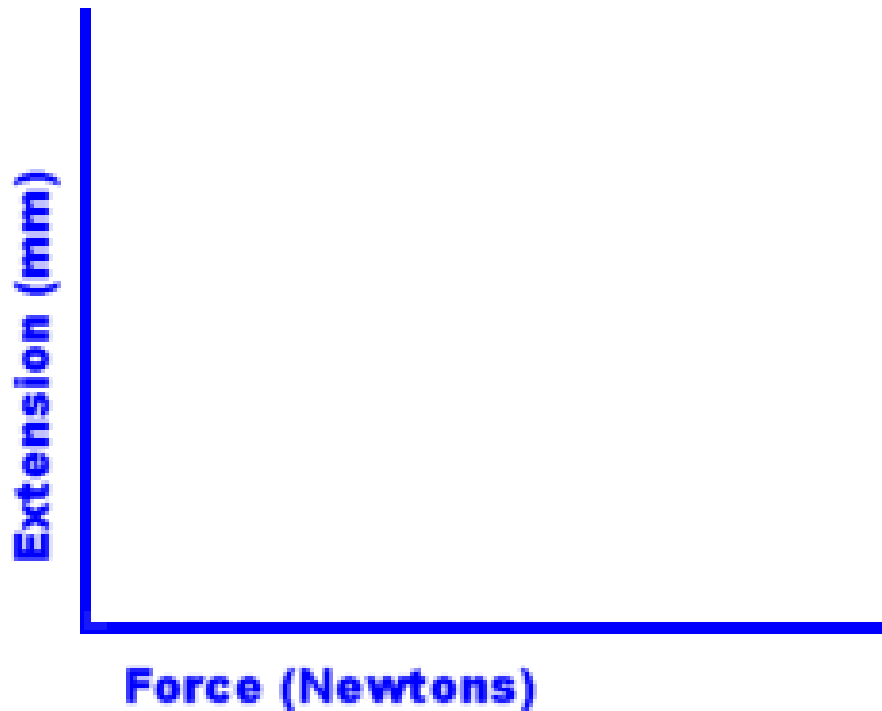


Load (force) (N)	Length (cm)	Extension (cm)
0.0	24.0	0.0
1.0	24.6	0.6
2.0	25.2	1.2
3.0	25.8	1.8
4.0	26.4	2.4
5.0	27.0	3.0
6.0	27.6	3.6
7.0	27.8	3.8
8.0	27.8	3.8
9.0	27.8	3.8

Law of Elasticity (Hook's law)



- Relationship between force and extension



Worked examples



1. A spring has a stiffness k of 20 N/m . what load is needed to produce an extension of 2.5 cm
2. A 75 kg person stands on a compression spring with spring constant 5000 N/m and nominal length of 0.25 m . What is the total length of the loaded spring?
3. What is the magnitude of the force required to stretch a 20 cm -long spring, with a spring constant of 100 N/m , to a length of 21 cm ?
4. What is the spring constant of a spring that needs a force of 3 N to be compressed from 40 cm to 35 cm ?

Uses of springs



- Weighing machines
- Pens
- Mattresses
- Bicycles
- Motorbikes
- Drives e.g in vehicles among others

Pressure



- Pressure can be defined as a force per unit area or
- The amount of force applied on a certain area.
- It is difficult to cut fruit with a blunt knife
- Surgical instruments have thin and sharp blades to reduce the amount of force applied to it on a patient. This ensures minimum damage is done when a patient is opened up
- The larger the surface area the less the pressure
- Vehicles are fitted with wide tyres to increase their surface area hence reducing the pressure

note



- Polar bears have large paws that act as snowshoes, spreading out the bear's weight as it moves over ice and snow
- If a force is spread out over a large area, its pressure is low
- If a force is concentrated to a small area, its pressure is high

Calculating pressure



- ***Pressure = force / area***

The SI unit of pressure is N/m^2 or Pa

Example

- Calculate the pressure exerted by a player weighing 600N standing on a single heel of area 1 cm^2 .*
- If the surface of the court is broken by pressures over 5 Million Pascal (5Mpa), will it be damaged?*
- Calculate the pressure produced by a force of 800 N acting on an area of 2.0 m^2*

Pressure in liquids



- Pressure in fluids act only upon the substance which are foreign to them
- Since a fluid has no definite shape, its pressure applies in all directions but does not act downwards

Note

- The deeper you go in water, the greater the pressure acting on you. This is due to the weight of water above you

Pressure, depth and density



- Pressure p is proportional to depth h .
- Pressure also depends on the density ρ of a material
- **$P = \rho h g$**
- ***Example***
- Calculate the pressure on the bottom of swimming pool 2.5m deep. How does the pressure compare with atmospheric pressure 100000 kg/m^3 ?

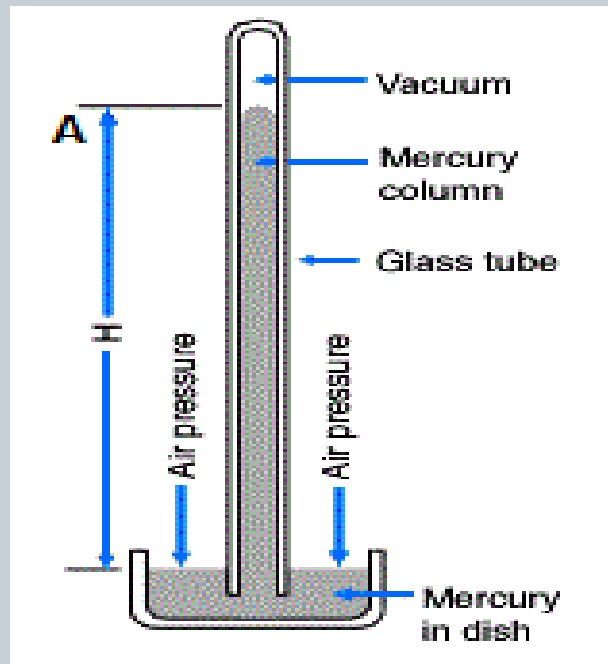
Instruments for measuring pressure



- Pressure is measured by an instrument called barometer
- There are two types of barometer, mercury barometer and aneroid barometer
- Difference in pressure is measured by manometer

Instruments

a) Mercury Barometer



b) Manometer

